

Pattern Positions

Time limit: 1.00 s

Memory limit: 512 MB

Given a string and patterns, find for each pattern the first position (1-indexed) where it appears in the string.

Input

The first input line has a string of length n .

The next input line has an integer k : the number of patterns. Finally, there are k lines that describe the patterns.

The string and the patterns consist of characters a–z.

Output

Print the first position for each pattern (or -1 if it does not appear at all).

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq k \leq 5 \cdot 10^5$
- the total length of the patterns is at most $5 \cdot 10^5$

Example

Input	Output
aybabbtu 3 bab abc a	3 -1 1